Human-Aware Waypoint Planner for Mobile Robot in Indoor Environments

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***Abstract*—As the utilization of robots in indoor environments increases, it has become common for humans and robots to co-exist in such environments. Most human-aware navigation algorithms only considered humans in the robot's field of view. However, in cases of L-shape corridors, there is a high possibility that human suddenly appears. To deal with this situation, we propose an improved corner detection algorithm and a novel waypoint planner, WPC. The proposed algorithm is validated through simulations using PedSim and Gazebo.**

Keywords—Human-Aware Robot Navigation, Human-Robot Interaction, Corner-Detection, Corner-Turning

I. Introduction

With the increase of contactless services and the development of mobile robots, mobile service robots are well used in indoor environments. Accordingly, robots commonly co-exist with people in hotels, hospitals [1], and airports. To navigate these dynamic human environments safely and smoothly, a robot should be aware of humans and understand the social rules in public places. Hence, many recent works have proposed various social navigation methods for mobile robots in dynamic human environments.

However, there are some regions where the robot's field of view is limited in the indoor environments, especially a corner of the corridor as shown in Fig.1. Since it is also true with humans, they tend to secure their view beyond a corner when passing through this area [2].

On the other hand, classical robot navigation [3], [4] mainly plans the shortest path to a goal with any exceptions for the occluded region. For this reason, when the robot navigates the corridor with a corner, the global path is planned along the wall. But the LiDAR or camera within a robot cannot observe over the corner. Thus, robots cannot prepare for obstacles beyond the corner or collisions with humans walking from the opposite side.

Likewise, if a robot suddenly emerges behind a corner, humans may feel embarrassed and have to change their trajectory for turning the corner abruptly. In this way, the classical navigation method makes the human move unnecessarily, that is, Human-Robot Interaction (HRI) is ignored. Therefore, in a corridor with such a blind corner, the robot needs a new path planning method for corner turning by recognizing a human appearance earlier. It maintains a safe distance between humans and robots and avoids unnecessary movement of humans.

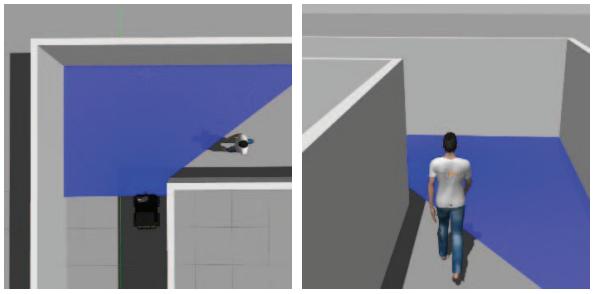

Fig. 1. A Robot and a human cannot see each other around a corner because the field of view is limited. (Left) View from a robot, (Right) View from a human. The blue zone represents the laser range within the robot. It means an area where a robot can see, so it cannot capture the human from the opposite side.

Mobile Service Robot automatically drives based on a map. We detect all corners of the map using our Corner Detection Algorithm that improved the existing corner detecting method on the image of the created map. Based on detected corners, we proposed an algorithm that generates a new path when a global path overlaps with a corner.

Finally, we build a realistic indoor simulation environment with an L-shape corner because safety is not guaranteed if humans and robots coexist. The human trajectory was generated based on the data analyzed in [2].

To summarize, our key contributions are:

(1) An approach that detects corners in a map

(2) A novel waypoint planner for a corner that avoids unnecessary movement of humans without real-time contour detection.

(3) We demonstrate that our approach can be more efficient than the shortest-path planning algorithm in a corner.

The rest of the paper is organized as follows. In Section Ⅱ, we discuss the related works. Further details on such approaches are presented in section Ⅲ. Section Ⅳ provides our experimental validations. We also present a comparison with one of the existing path-planning methods. Finally, Section Ⅴ concludes this work and provides directions for future research.

II. Related work

A. Navigation in an Indoor Environment

Mobile robots commonly use path planning methods such as Dijkstra [3] and A*[4], which plan the shortest path from

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the current location to its goal. However, these methods are efficient only in static environments because they do not consider human existence. So, in a crowded or dynamic environment where a human moves nearby a robot, there may be a "Freezing Robot Problem" which is the robot suddenly stops.

Meanwhile, humans understand spatial communication and follow social rules for human-human interaction. For example, if two people are walking toward each other in a narrow corridor, they both make efforts such as passing by the right to avoid a collision and to make the navigation safe and smooth. Likewise, for human-robot interaction (HRI) to match human-human interaction, spatial communication between humans and robots should not be ignored.

Therefore, navigation methods for HRI have been studied in various ways. [5] proposed the reactive planner, which provides a multi-robot navigation framework in human environments. With reactive-based planning, the robot changes its direction when a person appears in its way of it.

[6], [7] designed a probabilistic inference mechanism motivated by humans and used it to understand the environment and decide what future behaviors are adequate for it. Researchers in [8] enabled the robot to learn the social norms from the observed data and integrate them into its motion planning. Like this, with the predictive navigation framework, the mobile robot estimates the future states of the other agents and makes a reaction to avoid collision in advance.

As deep neural networks develop, many methods of navigating by learning robots with deep neural networks are also studied. [9], [10] using deep reinforcement learning to make robots not only behave like real humans but also avoid collisions with others. It is efficient because the learning proceeds with actual human trajectory data, so the robot can reproduce human behavior similarly.

However, most studies address only the visible humans in an open corridor and do not consider the possible emergence of humans that are not visible currently. Thus, researchers in [2] analyzed human behavior at the L-shape corner and proposed the robot's movement trajectory to avoid a collision. [11] introduced the concept of "invisible human" integrated into human-aware navigation and presented a methodology to estimate the locations of the invisible human. However, it requires contour detection for every laser scan data, which consumes the robot's computing resources.

B. Corner Detection and Corner Turning

In ROS[12], path planning is performed based on costmap [13]. For the robot to turn the corner at a certain radius, the inflation layer of the obstacle must be increased. However, if the inflation layer is increased for the whole static obstacle, the working area of the robot will be decreased. To overcome this problem, [14] detected corners from the static map using contour detection and convex hull algorithm. After detecting corners, they assigned additional cost only for the detected corners.

However, the corner detection of [14] need to run contour detection and convex hull algorithm several times, and the user of it need to know the exact hierarchy of the contour. Additionally, [14] cannot handle human-robot interaction. Suppose the robot has encountered human while moving alongside the assigned cost at the corner. In this case, the robot cannot use all the options to avoid human due the assigned cost. When the robot is turning the corner, all the options should be available for the robot.

III. Method

The proposed algorithm consists of two parts: corner detection and Waypoint Planner for Corners (WPC).

A. Corner Detection Algorithm

For the WPC to start its operation, the corners of the map must be detected. Our corner detecting algorithm is based on [14]. First, the map image needs to be loaded and the binary image is built from it. We use the same map data with ROS move_base stack. Then, the contour detection algorithm detects contour points and stores them in array A. Next, the outermost contour is determined using the contour area. Then the convex hull is built from this outermost contour, and its points are stored in an array B.

Since the points in B are the points of the map boundary, they must be removed from A. Thus, we calculate the distances between every point $a \in A$ and $b \in B$. If the distance is less than *dst_threshold*, a is removed from A. This sequence is shown in Fig. 2.

After this process, only the contour points inside the boundary remain in A. These points are corners that limit the field of view of the robot. Fig. 3 shows the result of our corner detecting algorithm.

B. Waypoint Planner for Corners: WPC

Fig. 4 shows the flowchart of WPC. WPC waits for the global path to be planned. Once the global path is planned, the corners of interest are chosen from A and stored in array C. For this, we calculate the distance between the global path and point $a \in A$. If a is close to the global path, a is copied to array C.

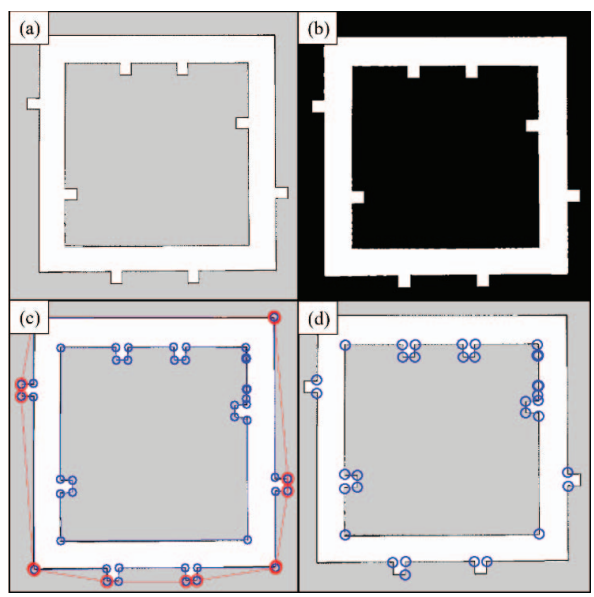

Fig. 2. The sequence of our improved corner detecting algorithm. (a) Loaded map image. (b) Binary image built from (a). (c) Convex hull (red) and contours (blue) marked on the map. (d) Detected corners of the map. Convex hulls are removed from contours.

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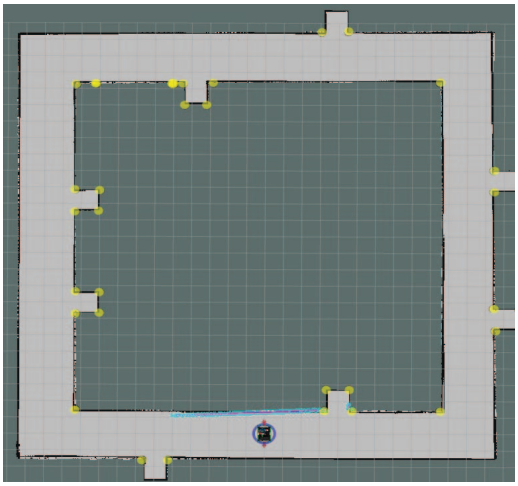

Fig. 3. Detected corners marked in RVIZ. The yellow markers indicate the location of the detected corners.

For each corner in *C*, we first find two points on the global path: Pin and Pout. Then, we interpolate between Pin and Pout using circular waypoints. The direction of waypoints is determined by $\Delta\theta$, which is the angular difference between Pin and the next point of it. This process is shown in Fig. 5.

After the waypoints are created, the robot starts its operation. When the robot gets closer to the corner, the WPC backs up the global goal and assigns a waypoint until there is no more

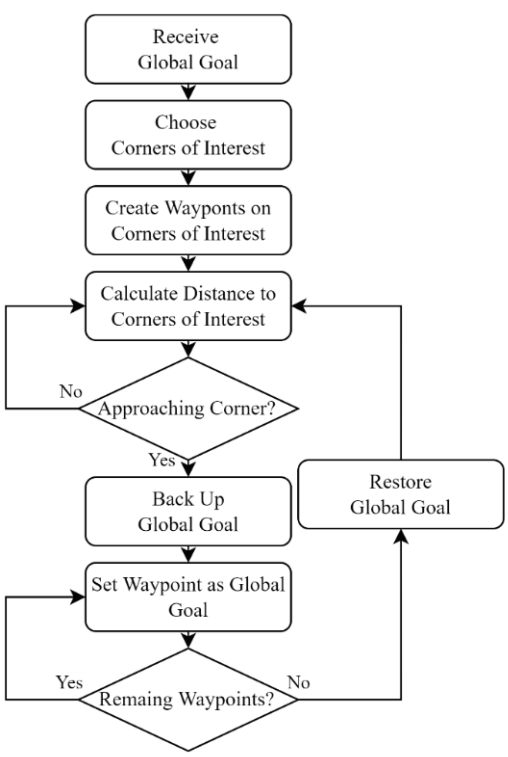

Fig. 4. Flowchart of WPC.

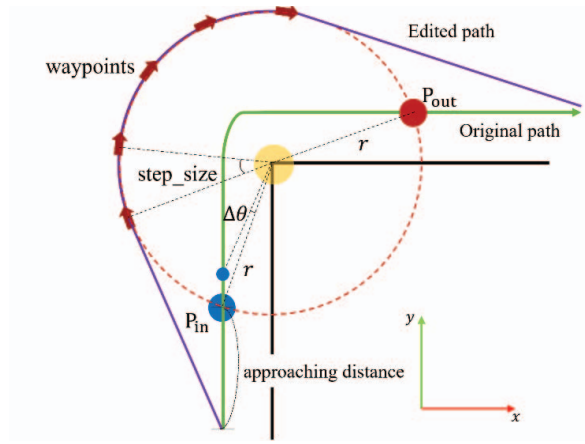

Fig. 5. Waypoint generation. The yellow circle indicates the detected corner, and the red arrows indicate the waypoint location and its direction. P_{in} and P_{out} are the points of the global path where the robot enters and exits the corner. *r* determines the distance between a robot and a corner. $\Delta\theta$ determines the direction of waypoints and *step_size* determines the distance between waypoints. *approaching distance* determines the distance for assigning waypoints to the move_base stack.

IV. Experiment

A. Experimental Setup

We use ROS-noetic with Ubuntu 20.04, and the scenario is simulated using Gazebo [15] simulator. Further, we also use a pedestrian simulator [16] for the human model based on Social Force Model[17]. The robot and a human encounter a corner in our scenario. The simulated human has a maximum speed of 1.4m/s like real humans and uses pedestrian trajectory data from [2]. The robot used in our experiment is Husky UR3 with LiDAR. We experimented shortest path method(move_base) and our method(WPC) 15 times each.

B. Experimental Evaluations

TABLE I. The evaluations of WPC against Move_base

Method	Metrics	
	Average Navigation time[s]	*Success rate [%]*
Move_base	39.76	86.67
WPC (ours)	40.00	100

First, we analyze the trajectory of the robot and the human in simulations. The PedSim is based on Helbing's social force model [17]. The social force model modeled human discomfort as a repulsive force, which appears to be a characteristic of keeping distance from obstacles in our simulations, for example, walls or other agents. Since we applied the repulsive force between robot and human, how much the human trajectory changes should be regarded as discomfort that humans experience. Therefore, we compare the trajectories obtained through the simulations.

As shown in Fig. 6(a), the robot using move_base turns corner following its shortest path to the goal. Thus, in Fig. 6(b), the trajectory of a human with the robot using move_base changes dramatically because the robot suddenly appears. In contrast, when WPC is used, a new path is created around the

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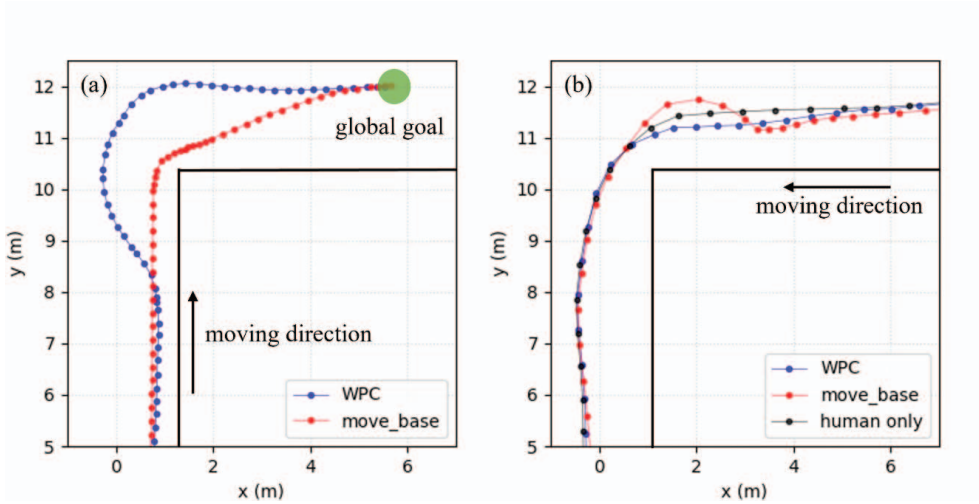

Fig. 6. Trajectories of the robot and human in simulation. Each markers indicates the location in 0.5 sec interval. (a) and (b) shows the trajectories of robot and human, respectively. The red trajectory indicates the trajectories when the move_base is used, and the blue trajectory indicates the trajectories when using WPC. In (b), the trajectory of the human without any robot is indicated in black.

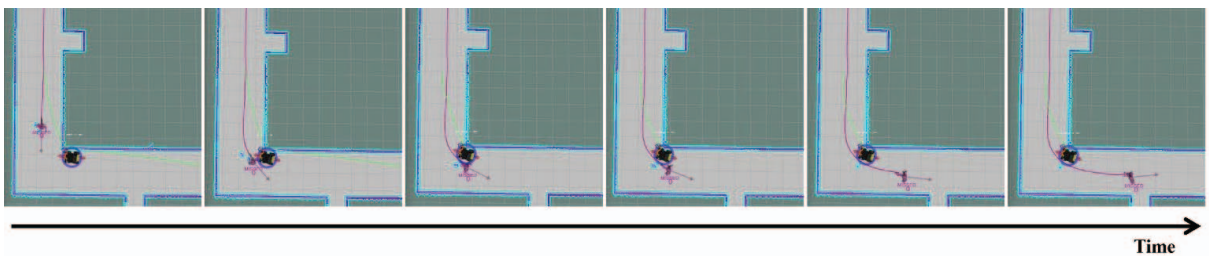

Fig. 7. With the shortest path planning method, the robot stops its navigation process because of the remaining cost. A human trajectory is marked in pink and the remaining cost of the passed human in blue.

COI, and the human trajectory stayed similar to the original one. Which means that humans can avoid robots by finding robots earlier without changing their routes. Quantitatively, the average speed of the human was 12.8 % higher than move_base, and the change of direction was 28.4 % lower than move_base. These results show that applying WPC will make humans feel less uncomfortable around the corner when the robot suddenly appears.

Next, we analyze the navigation performance of WPC. The success rate is the probability that the robot reaches the goal from its initial position, and the Average time is the average navigation time of the robot. As shown in Table 1, our waypoint method succeeded in all experiments, but the shortest path method failed twice.

Once a moving human is detected in the LiDAR, the human trajectory remains on the costmap even though a human passed completely. The robot recognizes the remaining cost on the costmap as a static obstacle and it re-plans the path or stops navigation as shown in Fig. 7. On the other hand, our method plans a new waypoint path based on the corner, considering a human may appear behind the corner. It showed a 100% success rate without a notable difference (0.6%) in navigation time from the shortest path method. This result clearly shows that our method operates more effectively than traditional path planning methods in an environment where humans and robots coexist.

However, there is a drawback in the aspect of energy. Whenever the waypoint is assigned, the value of /cmd_vel topic becomes 0. This causes a huge change in robot velocity. The linear acceleration showed 287.3% increment, and the angular acceleration showed 43.28% increment.

V. Conclusions and Future Work

Unlike existing robot navigation methods, mobile service robots in recent days should understand the social rules of humans in indoor environments where humans co-exist. Especially in a corridor with an L-shape corner, there is an occluded area beyond the corner.

To navigate these areas safely, this paper proposes a novel waypoint planner for a corner. Our waypoint planner demonstrates completely safe navigation than the shortest path method while keeping the navigation time similar. However, there are still some limitations to this approach. If several corners are overlapped, the robot cannot pass the area. Also, the robot consumes a lot of energy due to the speed change. The reason for these limitations is using waypoint-based move_base. It can be solved by expanding the path planner.

In future work, we will consider various scenarios, such as the doors where humans suddenly appear or an indoor crowded environment.

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